

Kshitij Chaturvedi

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EDUCATION

Guru Gobind Singh Indraprastha University

Nov 2021 – June 2025

- Bachelor of Technology in Electronics and Communication Engineering (GPA: 9.0/10.0)

EXPERIENCE

AI/ML Engineer, Le Quirks – SF, Bay Area (Remote)

Aug 2025 – Present

- Deployed **Rag-SQL Agent** on legacy Oracle (Azure), cutting query workload **50%+** & saving **40%** ops cost w/o migration.
- Built **retail demand forecasting** (XGBoost, LSTMs, GRUs); **signal decomposition** + XGBoost cut MAPE 15%.
- **Used:** NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, Tensorflow, SBERT, Chroma, Azure Cosmos DB, Qwen3, Hugging Face Transformers, PyTorch, LlamaIndex, LlamaParse, FastAPI, DVC, Azure Ai Foundry.

Data Analyst Intern, DLC Group of Companies – Dwarka, Delhi

Jan 2025 – April 2025

- Built **predictive models** (Python, SQL) to improve forecasts & cut query time by **30 mins daily**.
- Deployed **Adaptive-RAG** with re-ranking & hallucination grading, boosting query efficiency 60%+.
- **Used:** NumPy, Pandas, SciPy, Scikit-Learn, Statsmodels, Matplotlib, Seaborn, SBERT, FAISS, Llama3.2, Tavily, BeautifulSoup4, Hugging Face Transformers, PyTorch, LangChain ecosystem (LangChain, LangGraph, Ollama), FastAPI.

AI Prompt/Research Analyst Intern, Rau's IAS Study Circle – Rajendra Place, Delhi

July 2024 – Aug 2024

- Analyzed 20k+ weekly student records to identify dropout and retention trends across 15+ subjects.
- Revealed a 25% higher retention rate linked to faculty strategies, directly shaping curriculum planning.
- **Used:** SQL, Python, NumPy, Pandas, Scikit-Learn, TensorFlow, Statsmodels, Matplotlib, Seaborn.

SDE Intern, Radian Arc (Synthem) – Hyderabad (Remote)

Feb 2024 – July 2024

- Built multilingual AI health assistant, reducing computation by 35% through model distillation and batching.
- Led an 8-member team to deploy and optimize the system, reducing per-query costs by 40% with caching.
- **Used:** Python, SBERT, ChromaDB, Hugging Face Transformers, LangChain, PyTorch, OpenAI, Flask, React Native, Docker.

RESEARCH PROJECTS

How Signal Decomposition affects Cloudburst Prediction?

[GitHub Repository](#)

- Designed a **time-series forecasting** pipeline using CNN-1D, LSTM, GRU, and TFT with **EEMD decomposition + FCR reconstruction**, achieving best results via ensemble modeling.

Adaptive RAG for Enterprise Support Systems, with Budgeted Compute

[GitHub Repository](#)

- Built a **RAG pipeline with SLMs** (28% lower latency) enhanced by an **agentic query rewriter** and **multi-vector hybrid search router** (keyword + semantic, RRF) for improved retrieval.
- Processed **1K+ docs & 100+ queries** by handling **multi-modal data** (text, tables, images) with OCR layout detection.

SKILLS

Languages: Python, SQL, JavaScript

Concepts: OOPs, DBMS, Data Analytics & Visualization, ML (Supervised, Unsupervised), Deep Learning, Natural Language Processing, Computer Vision, Video Analytics, LLMs, GenAI, Transfer Learning

Libraries: NumPy, Pandas, Scipy, Statsmodels, Scikit-Learn, TensorFlow, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, PaddleOCR, OpenCV, Supervision, Flask, Selenium, BeautifulSoup4, PyTest

Technologies: MySQL, Git, GitHub, Postman, Docker, Azure, DVC, Ollama, Overleaf, Next.js, Tailwind CSS

ACHIEVEMENTS

- Advised **6+ startups** on ML workflows, cutting processing time **30%** and boosting accuracy **15%**.
- Guided AI adoption strategies, enabling efficient scaling and **\$500K+ revenue growth in 2 months**.
- **Judged** BVP Hex Hackathon (Nov '24); **Mentored** 5+ hackathons across Delhi.
- **Qualified SIH 2023 & 2024** under ML and Climate tracks.
- **Top 10** in HackMait'22 (MAIT) & EmpowHer Hack (IEEE-IIITD); **Top 30** in Vihan 2.0 (IEEE-DTU'23).
- Held leadership roles as **Management Head** at ECELL-ADGITM; **Instrumental Head & Treasurer** at Swaranjali-ADGITM.

CERTIFICATIONS

- **Machine Learning Specialization I - III** (Supervised, Advanced, Unsupervised) by **DeepLearning.AI (Andrew Ng)**
- **IBM Certified Artificial Intelligence Analyst**